

HoloArm: A Face-Following 3D Display Using Autostereoscopic Display and Robot Arm

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Figure 1: Our *HoloArm* concept allows for a unique autostereoscopic experience at various viewing angles. The viewer can (a) face forward, (b) move 20 degrees to the left, or (c) move 30 cm up, 20 cm right, 20 degrees up, to look up

Abstract

In this demonstration, we propose the *HoloArm*, an actuated autostereoscopic display system that constantly maintains the view in front of the user. Three-dimensional displays aid spatial awareness and increase immersion in the content. As a common display type, autostereoscopic displays attract attention as they do not require users extra gears to watch 3D content. However, autostereoscopic displays have challenges in limited viewing angles and a viewing area dependent on the display size. We thus propose a face-following autostereoscopic display, the *HoloArm*, which dynamically adjusts the autostereoscopic display according to the user's face orientation. It provides a wide viewing angle and large viewing area without requiring users to wear additional equipment such as glasses. Through the demonstration with the *HoloArm*, we provide an experience to the visitors as if they have a dynamic window into the virtual world, providing the user with a deep immersion as if looking into another world.

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CCS Concepts

• Human-centered computing → Virtual reality; Interaction design theory, concepts and paradigms.

Keywords

Autostereoscopic display, virtual reality, telepresence, robot arm

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1 Introduction and Background

Three-dimensional (3D) display technology [5, 6, 8, 14] has long captivated researchers and continues to be a fascinating field of study. With the recent rise in virtual reality (VR) and augmented reality (AR), 3D display technology is finding applications in a variety of fields, including medicine [10], education [11], and entertainment [12]. By providing depth information, 3D displays can facilitate the user's grasp of space better than 2D displays [9].

Among various types of 3D displays, autostereoscopic (glasses-free) displays are particularly convenient since they do not require users to wear special equipment such as 3D glasses or head-mounted displays (HMDs) [3]. This removes the hassle of putting on and taking off devices, making this technology attractive for a wide range of scenarios.

Nevertheless, existing autostereoscopic displays face the challenge of having a narrow viewing angle and viewing range [1]. While several studies have attempted to expand the viewing angle through the use of optical systems and eye-tracking approaches [7, 13], the viewing angle and viewing area still limited since it depends on the detectable range of the sensor.

We thus propose a new display system, *HoloArm*, that mounts an autostereoscopic display on a robotic arm and autonomously tracks the user's gaze position to present 3D images. *HoloArm* offers a broader viewing angle and viewing zone by following the user's gaze with autostereoscopic vision, functioning just like a transparent window from the real world to virtual world objects. Users gain a sense of 3D space existing before their eyes, obtaining a more extensive and more unrestricted perspective into a virtual world (Fig. 1).

2 System Concept

Fig. 2 illustrates the concept of the *HoloArm*. *HoloArm* uses a robot arm to dynamically move the autostereoscopic display, enabling users to see the virtual world with a wide viewing angle and field of view. The robot arm always moves to face the user's face position, and the projected image dynamically changes according to the movement of the display. The user feels the autostereoscopic display as a window to the virtual world since the virtual world expands in accordance with his/her movements. As shown in Fig. 2, the *HoloArm* can also display real-world scenes by streaming images from a stereo camera placed remotely, enabling near life-size telepresence.

3 Implementation

We have implemented a prototype (Fig. 3) to demonstrate the concept of the *HoloArm*. In this prototype, a robot arm is fixed to the base, and an autostereoscopic display is attached to the end of the arm with a standard VESA mount. The user's face is tracked via an external web camera. The 6DoF information of the user's face is sent to the system, which in turn computes the desired end-effector position of the arm. The robot arm moves the display to follow the line of sight based on the transmitted information. The image displayed by the display moves by the displacement of the display. After explaining the hardware implementation (Sec. 3.1), the tracking details (Sec. 3.2) and the arm serving details (Sec. 3.3) are explained.

3.1 Hardware Setup

We used a Sony ELF-SR2 (27-inch) for the autostereoscopic display. It uses an eye-tracking camera to detect the human line of sight and displays stereoscopic images in a lenticular format according to the position and rotation of the human eye. It is compatible with Unity3D, and the displayed images and applications are managed by Unity3D. We used a Dobot CR10 for the robot arm, of which

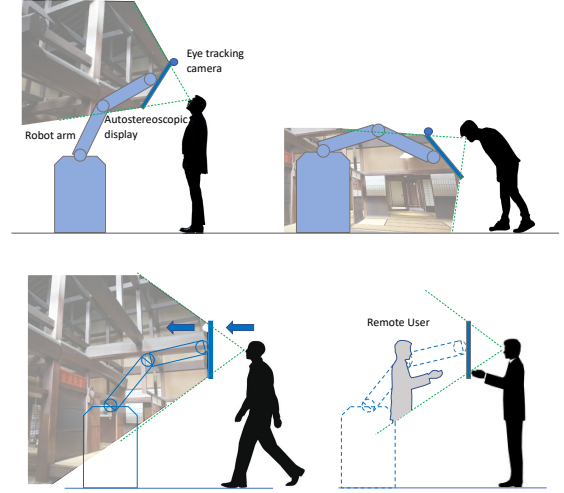


Figure 2: A schematic illustration of our *HoloArm* concept: (Top) The autostereoscopic display always moves directly opposite the user's face. (Bottom left) The entire display system moves with the user's movements. (Bottom right) Life-size communication with a remote user is also possible.

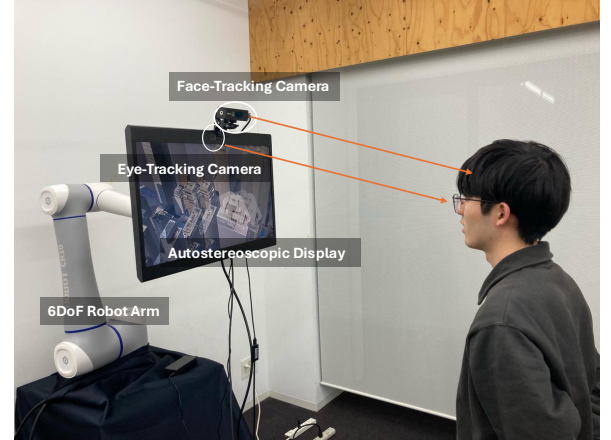


Figure 3: Actual *HoloArm* photograph. The autostereoscopic display is mounted on a 6-axis robot arm. The user is tracked using two tracking mechanisms: the eye-tracking camera that comes standard with the stereoscopic display and the face-tracking camera that is attached externally.

the payload is 10 kg, as the Sony ELF-SR2 weighs approximately 8.2 kg. Additionally, we use ASUS Xtion2 (RGB+Depth) camera as an eye-tracking camera. The viewing angle of this camera is 74° horizontal, 52° vertical, and 90° diagonal. By using an external web camera, we get more precise data about the user's face location and rotation, which also gets sent to Unity3D. In Unity3D, we controlled the robot arm via its TCP/IP protocol, which allowed us to set and

monitor the 6DoF pose information from the arm's base and guide the arm's tip to the desired position.

3.2 Capturing and Following User

As illustrated in Fig. 3, the system employs two distinct mechanisms to track the user. One is a face-tracking camera attached to the display. We use OpenSeeFace[4], developed by Emiliana, to capture the translation and rotation of the user's face. The robot arm then moves according to this face-position data. The second mechanism is an eye-tracking camera built into the ELF SR2 (autostereoscopic display). This camera tracks the user's eyes and renders a 3D image in real time.

By combining these two cameras, the system maintains robust, uninterrupted tracking. Moreover, as long as the user's face remains within the eye-tracking camera's tracking range, that camera can continue to track and render the 3D view at high speed, independently of the robot arm. This means the eye-tracking camera can update the display before the robot arm reaches just in front of users, thereby minimizing any perceived delay from the user's perspective.

3.3 Arm Servoing

The arm is controlled using an API provided by Dobot. This API issues movement commands to the robot arm at 33 Hz. To prevent the arm from rattling around the same place at high speed, it is filtered by OneEuroFilter [2], an adaptive low-pass filter that reduces jitter while maintaining responsiveness by adjusting its cutoff frequency based on movement speed, developed by DarioMazzanti, and moves above a threshold value are ignored. This allows the robot arm to perform smooth movements. To achieve even smoother motion, linear completion and smoothing mechanisms are also applied. Specifically, linear completion moves the arm to 50% of the target position every frame, and smoothing is set to use the average of the last five frames as the target value. This allows the robot to smoothly follow the user's movements while filtering out sudden abnormal values caused by tracking errors, thereby improving stability and safety.

4 Demo Experience

Our demonstration consists of three main experiences that showcase the unique capabilities of *HoloArm's* face-following display system.

4.1 Immersive Virtual Object Viewing

The *HoloArm* enables an immersive virtual object viewing experience that could not be achieved with traditional displays. The system displays cultural artifacts such as medical images, CT scans, and sculptural works in real size.

HoloArm also provides an immersive experience in a spectacular virtual space. Users can explore the interiors of majestic museums and historic buildings and view their surroundings from within those spaces, and the *HoloArm's* unique autostereoscopic viewing and wide viewing angles provide a deeply immersive experience without the need to wear a device. The system follows the user's movements, creating an unprecedented experience, as if a magic window were floating in space, allowing the user to peer through it

into the virtual world beyond. This intuitive and natural observation experience opens up new possibilities that could not be realized with conventional display technology.

4.2 Dynamic Telepresence

A telepresence experience can be created by combining the *HoloArm* with a stereo camera placed at a distance; by projecting the stereo camera image onto the *HoloArm*, a real world image can be projected. As the user moves around, the *HoloArm* automatically follows his or her movements, creating an experience that is like an open window to the remote location.

Notably, the system enables intuitive communication between remote and local users. When the remote user points, "Look here!" the robotic arm automatically moves in that direction, naturally guiding the local user's gaze. This enables communication with a shared spatial context, which is difficult to achieve with traditional videoconferencing.

The system maintains an optimal viewing angle for the user's perspective at all times, and combined with the stereoscopic view provided by the stereo camera, the remote space is experienced more naturally and immersively. The system provides the feeling of actually being there, breaking down the physical barriers of distance in remote communication. With *HoloArm*, switching between projected images (real world and virtual world) can be done quickly

4.3 3D modeling

In this application, participants engage in 3D modeling using a standard monitor and verify the completed model using the *HoloArm*. Pressing the Unity Run button on the standard monitor activates the Unity scene displayed through the *HoloArm*. Users can move around the *HoloArm* to examine detailed aspects of the model that are typically less visible on a conventional monitor.

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